

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
BELAGAVI – 590 018, KARNATAKA



Shri Bhagwan Mahaveer Jain Educational & Cultural Trust ®
JAIN COLLEGE OF ENGINEERING, BELAGAVI



**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

PRE-FINAL YEAR
(2025 – 2026)

MINI-PROJECT REPORT

On

"SMART ASTHMA KIT"

PROJECT GUIDE

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**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

CERTIFICATE

This is to certify that the **Mini-Project Work** entitled “**SMART ASTHMA KIT**” carried out by **Siddharth K. (2JI23EC140)**, **Trupti S. (2JI23EC165)**, **Shyambhavi N. (2JI23EC137)** and **Vikas G. (2JI23EC173)** are bonafide students of **Department of Electronics and Communication Engineering, Jain College of Engineering, Belagavi**, in partial fulfilment for the award of **Bachelor of Engineering** of the **Visvesvaraya Technological University, Belagavi** during the academic year **2024-2025**. It is certified that all corrections/suggestions indicated for project assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering degree.

Dr.Krupa Rasane

Project Guide

Dept. of E&CE, JCE Belagavi

Prof. V.R.Bagali

HOD, Dept. of E&CE

JCE, Belagavi

Dr. J. Shivakumar

Principal & Director

JCE, Belagavi

DECLARATION

We, **Siddharth K. (2JI23EC140), Trupti S. (2JI23EC165), Shyambhavi N. (2JI23EC137) and Vikas G. (2JI23EC173)**, students of 5th semester **B.E. Electronics & Communication Engineering, Jain College of Engineering, Belagavi** hereby declare that the dissertation entitled “**SMART ASTHMA KIT**” has been carried out in a batch and submitted in the partial fulfilment of the requirement for the award of Bachelor’s Degree in Electronics & Communication Engineering under Visvesvaraya Technological University, Belagavi during the academic year **2025– 26**.

<u>Name</u>	<u>USN</u>	<u>Signature</u>
Siddharth Kshatriya	2JI23EC140	_____
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Place : Belagavi

Date : 11/12/2025



Jain College of Engineering, Belagavi
Dept. of Electronics and Communication Engineering

VISION

To achieve excellence in education and research for developing globally competent, ethically sound electronic and communication engineers

MISSION

1. To provide a conducive environment for learning through structured student centric, teaching learning process
2. To nurture needs of society by infusing scientific temper in students and to grow as a centre of excellence with efficient industry-institute interaction
3. To inculcate self learning skills, entrepreneurial ability and professional ethics.

Program Educational Objectives (PEO's)

Graduates will be able to

- Contemplate real-time social problems and deliver efficient solutions
- Lead and succeed in professional careers
- Contribute through research and entrepreneurship

Program Specific Outcomes (PSO's)

Graduates in the UG program in Electronics and communication engineering will be able to

- Design, verify and develop analog and digital systems by using state of art technology to contribute to the societal needs.
- Apply the knowledge in various domains of IoT, real time systems, communication systems, VLSI and embedded systems, image and signal processing using hardware and software tools.



PROGRAM OUTCOME'S (PO'S)

Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the **engineering and management principles and apply these to one's own work, as a member and leader** in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change



Jain College of Engineering, Belagavi

Dept. of Electronics and Communication Engineering

Course Outcomes (CO's)

Course Outcome	Course Outcome Statement	Bloom's Level
BEC586.01	Design and develop an electronic system using modern tools by following professional ethics.	L3,L4
BEC586.02	Develop managerial, teamwork and communication skills.	L3
BEC586.03	Understand the design factors to be considered that impact society.	L2,L3

Bloom's Cognitive Levels

L1: Remembering; L2: Understanding; L3: Applying; L4: Analyzing; L5: Evaluating; L6: Creating

STRENGTH OF CO MAPPING TO PO/PSOS WITH JUSTIFICATION:

1. Slight (Low) 2. Moderate (Medium) 3. Substantial (High)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	1	3	-	-	2	-	-	-	2	2	2
CO2	-	-	-	-	-	-	-	-	3	3	2	-	-	-
CO3	-	-	-	-	-	2	2	-	-	-	-	-	-	-

CO – PO - PSO	Justification
CO1 → PO1 (3) CO1 → PO2 (3) CO1 → PO3 (3) CO1 → PO4 (1) CO1 → PO5 (3) CO1 → PO8 (2) CO1 → PO12 (2) CO1 → PSO1(2) CO1 → PSO2(2)	- Students apply the electronics and communication engineering skills and design products to solve complex engineering problems. - Students gain the knowledge of modern tools and skill usage. - Students can use research-based knowledge and methods including design of experiments, analysis and interpretation to provide valid conclusions. - Students can work ethically and professionally in the industry. - Students can design their projects in analog or digital fields. - Students can choose any technology to build their projects.
CO2 → PO9 (3) CO2 → PO10 (3) CO2 → PO11 (2)	- Students can effectively act as an individual, and as a member or leader and work in a team. - Students can comprehend and write effective reports and make documentation with effective presentation. - Learn to management and financial skills required for the execution of project.
CO3 → PO6 (3) CO3 → PO7 (3)	- Students can able to understand the impact of professional engineering solutions in societal and environmental contexts. - Students can be able to apply reasoning of contextual knowledge to assess societal, health and safety issues.

ACKNOWLEDGEMENT

Although a single sentence hardly suffices, we would like to thank almighty God for blessing us with his grace and taking our endeavour to a successful culmination.

We express our gratitude to our guide **Dr. Krupa Rasane, Professor, Dept. of E&CE, JCE, Belagavi**, for his valuable guidance and continual encouragement and assistance throughout the project work. We greatly appreciate the freedom and collegial respect. We are grateful to him for discussions about the technical matters and suggestions concerned to our topic.

We extend our sense of gratitude to **Prof. Shivanand Channi, Mini-Project Coordinator, Dept. of E&CE, JCE, Belagavi**, for extending support and cooperation which helped us in completion of the project work.

We extend our sense of gratitude to **Prof.V.R.Bagali , Head, Dept. of E&CE, JCE, Belagavi**, for extending support and cooperation which helped us in completion of the project work.

We would like to express our sincere thanks to **Dr. J. Shivakumar, Principal, JCE Belagavi**, for extending support and cooperation which helped us in the completion of the project work.

We would like to extend our gratitude to all staff of **Department of Electronics and Communication Engineering** for the help and support rendered to us.

We would like to extend our gratitude to all our family members and friends especially for their advice and moral support.

ABSTRACT

The Smart Asthma Kit is an IoT-based health monitoring system designed to assist asthma patients by providing continuous, real-time tracking of respiratory and environmental parameters. The project integrates two devices: a **wearable wristband** and a **spirometer**. The wristband monitors SpO₂, heart rate, temperature, humidity, and air quality using sensors such as the MAX30102 and BME680. The spirometer measures airflow and breathing performance using the MPX5010DP differential pressure sensor.

Both devices are powered by ESP32 microcontrollers and communicate with a web application through WiFi. The app displays live data, stores historical records, and provides alerts when abnormal readings or asthma-triggering environmental conditions are detected.

This system enables early detection of asthma triggers, improves patient safety, and offers a portable, convenient solution for long-term asthma management. The integration of hardware sensors, BLE / WiFi communication, and a user-friendly mobile interface makes the Smart Asthma Kit an effective tool for personalized respiratory health monitoring.

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CHAPTER 1

INTRODUCTION

1.1 MOTIVATION

Asthma is a chronic respiratory condition that affects millions of people worldwide, and many patients experience sudden attacks due to undetected triggers or improper monitoring of their respiratory health. Environmental factors such as poor air quality, humidity variations, temperature changes, and exposure to harmful gases significantly influence asthma symptoms. In many cases, patients are unaware of these changes, leading to unexpected emergencies. Traditional monitoring methods are not portable, do not provide continuous data, and rely heavily on manual observation. There is a strong need for a system that can monitor both respiratory parameters and environmental conditions in real time. The Smart Asthma Kit aims to fulfil this need by integrating a wearable wristband and a portable spirometer to help users manage their condition effectively, prevent severe episodes, and gain timely alerts to avoid critical situations

1.2 PROBLEM STATEMENT

Asthma patients often face emergencies because they are unable to monitor changes in their respiratory condition and surrounding environment in real time. Existing systems are not portable and do not provide continuous updates on vital parameters or asthma-triggering factors. There is a need for a compact solution that can track breathing performance and environmental conditions and notify the user immediately when unsafe situations arise.

1.3 OBJECTIVE OF PROJECT

- The objective of this project is to design a Smart Asthma Kit that combines a wristband and a spirometer to monitor vital respiratory and environmental parameters. It aims to provide continuous tracking of heart rate, SpO₂, air quality, temperature, and lung airflow in real time. The system helps patients detect early signs of asthma triggers. It improves safety by offering timely alerts through a mobile application.

- The project also aims to create a portable and easy-to-use solution that allows asthma patients to manage their condition effectively. By integrating wireless communication and a user-friendly app, the system supports better health monitoring. It ensures reliable data access for users at any time. This helps in preventing sudden asthma attacks and reducing emergency situations.

1.4 DESIGN APPROACH

Firmware Design & Development

- Requirement gathering for wristband and spirometer features such as heart rate, SpO₂, air quality, temperature, humidity, and airflow monitoring.
- Preparation of the software design document defining sensor connections, data flow, and BLE communication between devices and the mobile app.
- Firmware design and development for ESP32 XIAO (wristband) and ESP32 (spirometer) to read and process sensor data.
- Coding for Bluetooth Low Energy (BLE) transmission and ensuring smooth communication with the website.

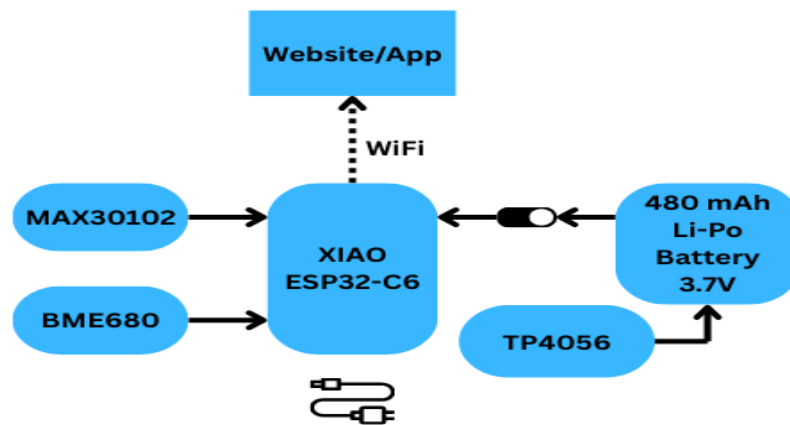
Testing

- Module testing of individual hardware components: MAX30102, BME680, MPX5010DP, ESP32 XIAO, Arduino uno R3 and battery module.
- Unit & integration testing of firmware with sensors and BLE communication between wristband, spirometer, and the mobile app.
- System testing to verify accurate data monitoring, real-time updates in the app, and proper alert generation under abnormal conditions.

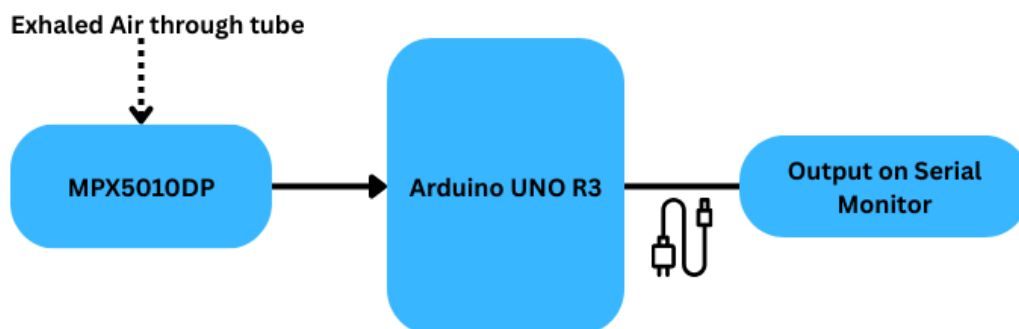
CHAPTER 2

HARDWARE DESIGN

2.1 BLOCK DIAGRAM



Wristband Block Diagram



Spirometer Block Diagram

EXPLANATION OF EACH BLOCK WITH PROPER NAMES AND THEIR FEATURES AS FOLLOW

2.2 WRISTBAND HARDWARE

2.2.1 XIAO ESP32-C6



Figure 1: XIAO ESP32-C6

The XIAO ESP32-C6 is a compact microcontroller designed for wearable and IoT applications. It supports Wi-Fi 6 and BLE 5.0, making it suitable for real-time wireless monitoring.

FEATURES

- Supports Wi-Fi 6 and Bluetooth Low Energy (BLE 5.0)
- Low-power RISC-V processor suitable for battery-operated devices
- Small form factor with multiple GPIO pins for sensor interfacing

XIAO ESP32-C6 PIN DESCRIPTION

SL NO	PIN	DESCRIPTION
1	3V3	3.3V Power Supply
2	GND	Ground
3	D0	Digital I/O
4	D1	Digital I/O / I2C SDA
5	D2	Digital I/O / I2C SCL
6	D3	Digital I/O
7	D4	Digital I/O
8	D5	Digital I/O
9	D6	Digital I/O
10	D7	Digital I/O
11	D8	Digital / Analog Input
12	D9	Digital / Analog Input

2.3 MAX30102

2.3.1 MAX30102 details

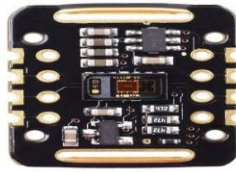


Figure 2: MAX30102

The MAX30102 is an integrated pulse oximetry and heart-rate sensor that uses LEDs and a photodetector to measure SpO₂ and pulse rate. Its compact size makes it suitable for wearable health-monitoring devices.

FEATURES

- Measures heart rate and SpO₂ using red and IR LEDs
- Low-power operation suitable for battery-powered systems
- I2C communication interface for easy microcontroller integration

MAX30102 PIN DESCRIPTION:

SL NO	PIN	DESCRIPTION
1	VIN	Power supply (3.3V / 5V)
2	GND	Ground
3	SCL	I2C Clock Input
4	SDA	I2C Data Input
5	INT	Interrupt Output
6	IRD	IR LED Control (Internal)

2.4 BME680

2.4.1 BME680 details

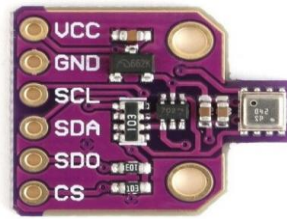


Figure 3: BME680

The BME680 is a compact environmental sensor capable of measuring temperature, humidity, pressure, and air quality from a single module. It is designed for portable and low-power applications such as wearable monitoring systems.

FEATURES

- Measures temperature, humidity, pressure, and gas (air quality)
- Low-power operation suitable for continuous monitoring system
- I2C interface for easy integration with microcontrollers

BME680 PIN DESCRIPTION:

SL NO	PIN	DESCRIPTION
1	VIN	Power supply (3.3V / 5V)
2	GND	Ground
3	SCL	I2C Clock Input
4	SDA	I2C Data Input
5	CS	Chip select (SPI,,Optional)
6	SD0	SPI Data Output (optional)

2.5 BATTERY AND CHARGING MODULE

2.5.1 BATTERY SPECIFICATIONS



Figure 4: Li-Po battery

The 3.7V, 480 mAh Lithium-Polymer (Li-Po) battery is used as the primary power source for the wristband module. It is lightweight and compact, making it suitable for wearable and portable devices that require continuous operation.

SPECIFICATIONS

- Battery Type: Lithium-Polymer (Li-Po)
- Rated Voltage: 3.7V
- Capacity: 480 mAh
- Charging Voltage: 4.2V (max)
- Discharge Current: Suitable for low-power IoT devices
- Weight: Lightweight, ideal for wearable applications

2.5.2 CHARGING MODULE

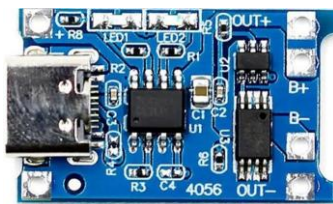


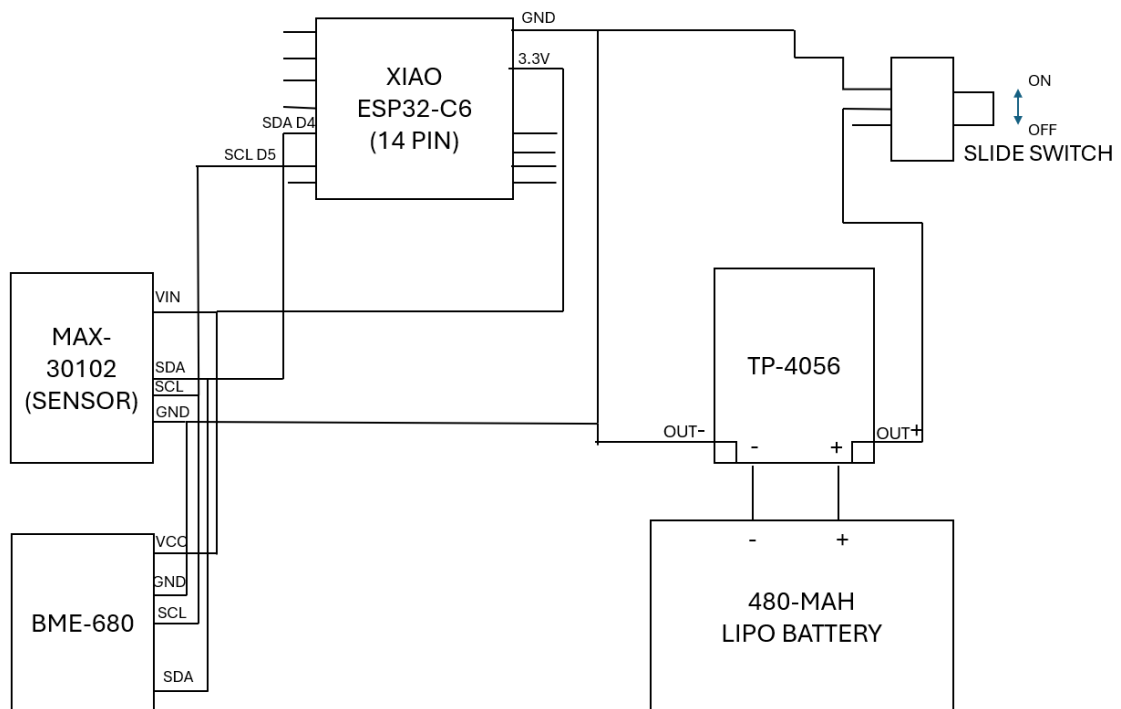
Figure 5: TP4056

The TP4056 is a linear Li-Ion/Li-Po battery charging module designed for safely charging 3.7V Li-Po cells. It provides a constant-current and constant-voltage charging method, making it ideal for compact battery-powered devices.

FEATURES

- Supports 1-cell 3.7V Li-Po battery charging with built-in protection
- Includes onboard micro-USB/Type-C input for easy charging

2.6 WRISTBAND INTERFACING



2.7 SPIROMETER HARDWARE

2.7.1 ARDUINO UNO R3



Figure 6: Arduino Uno R3

The Arduino Uno R3 is an ATmega328P-based microcontroller board widely used for prototyping embedded systems. It offers stable performance, multiple digital/analog pins, and easy programming, making it suitable for spirometer data acquisition and sensor interfacing.

FEATURES

- Based on ATmega328P microcontroller with 14 digital and 6 analog I/O pins
- Supports USB programming and communication for easy development
- Operates at 5V with built-in voltage regulator and crystal oscillator

Arduino Uno R3 PIN DESCRIPTION:

SL NO	PIN	DESCRIPTION
1	5V	Regulated 5V Output
2	3.3V	Regulated 3.3V Output
3	GND	Ground Pins
4	VIN	External Power Input
5	A0-A5	Analog Input Pins
6	D0(RX)	UART Receive
7	D1(RX)	UART Transmit
8	D2-D13	Digital I/O Pins
9	DPWM Pins	D3, D5, D6, D9, D10, D11
10	RESET	Reset the Microcontroller

2.8 MPX5010DP

2.8.1 MPX5010DP DETAILS

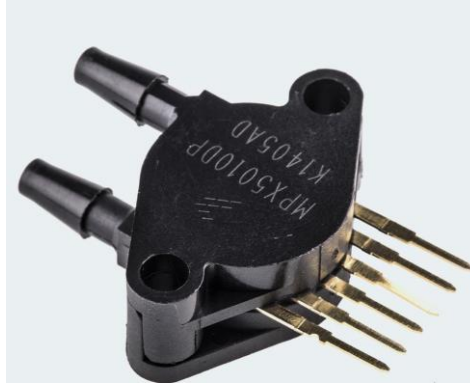


Figure 7: MPX5010DP

The MPX5010DP is a differential pressure sensor that provides an accurate and linear voltage output proportional to the applied pressure. It is suitable for airflow measurement applications such as spirometers.

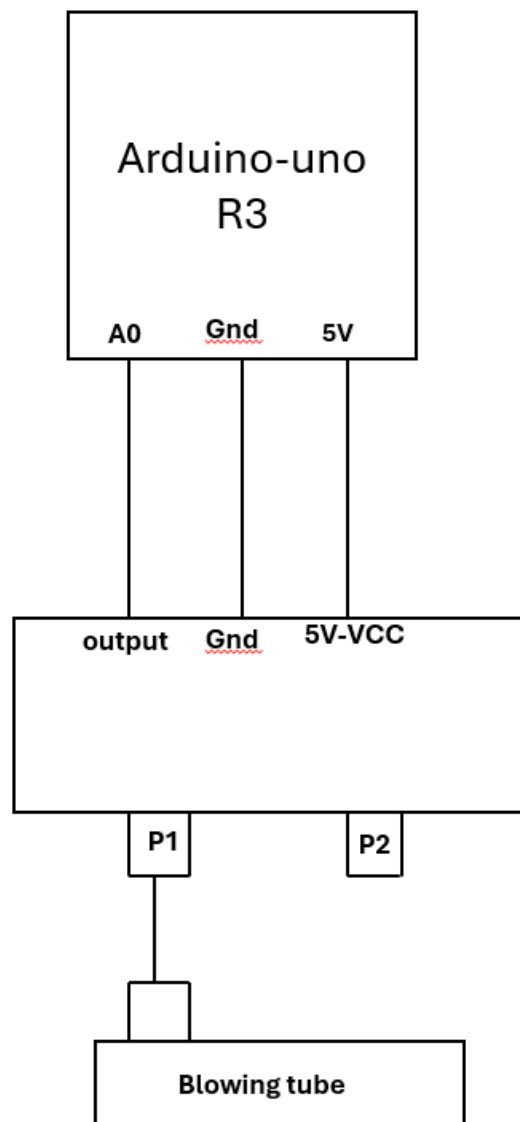
FEATURES

- Provides accurate differential pressure measurement with linear analog output
- Low-pressure sensing range suitable for detecting human breathing airflow
- Easy interfacing with microcontrollers through its analog voltage output

MPX5010DP PIN DESCRIPTION:

SL NO	PIN	DESCRIPTION
1	VCC	Power Supply (5V)
2	GND	Ground
3	VOUT	Analog Output Voltage
4	NC	Not Connected
5	NC	Not Connected
6	NC	Not Connected

2.9 SPIROMETER INTERFACING



CHAPTER 3

SOFTWARE DESIGN

3.1 DEVELOPMENT TOOLS

The software development of this project uses a combination of tools for programming, compiling, uploading, and visualizing sensor data. These tools support the ESP32 microcontroller, enable real-time data transfer through Wi-Fi, and provide a web-based dashboard for monitoring asthma-related parameters.

3.2 ARDUINO IDE

Arduino IDE is used to write, compile, and upload the firmware code to the XIAO ESP32-C6 and Arduino UNO. It provides an easy-to-use programming environment with built-in libraries for the MAX30102 sensor, BME680 sensor, MPX5010DP pressure sensor, and Wi-Fi communication. The IDE also supports serial monitoring for debugging and real-time testing of sensor outputs.

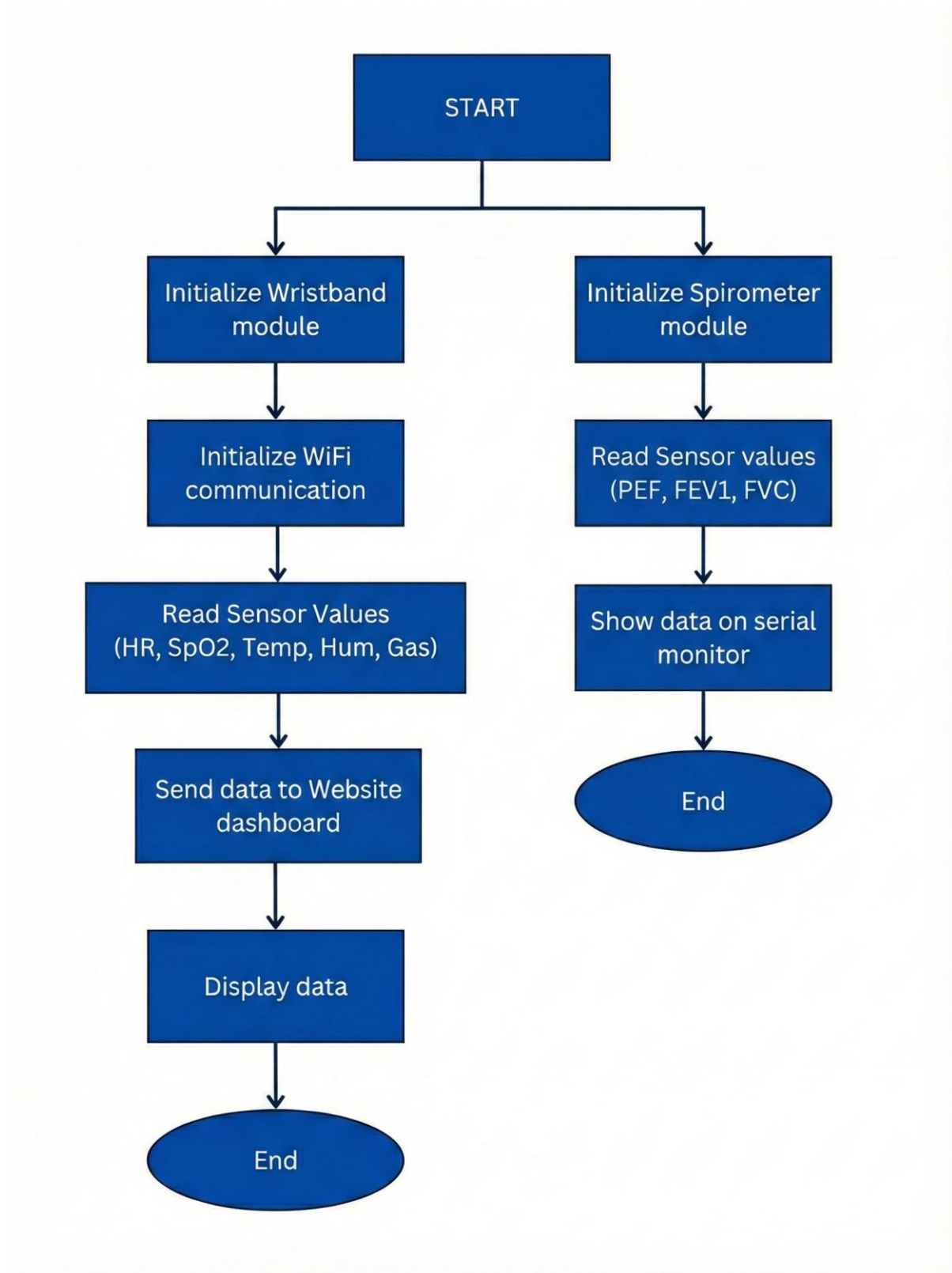
3.3 ESP32 WEB SERVER

The ESP32 Web Server is implemented to transmit the collected wristband and spirometer sensor data over Wi-Fi to a web-based dashboard. The ESP32 hosts or communicates with a server using HTTP/MQTT protocols, allowing data to be displayed in real time on any internet-connected device. This method replaces the need for a mobile application and provides an efficient interface for monitoring, visualization, and alert generation.

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3.4 DESIGN

3.4.1 MAIN FLOW DIAGRAM



CHAPTER 4

APPLICATIONS

APPLICATIONS

1. Personal Health Monitoring:

Continuous monitoring for asthma patients at home or outdoors.

2. Hospitals and Clinics:

Useful for patient observation and pulmonary function assessment.

3. Telemedicine & Remote Healthcare:

Doctors can receive patient data for remote consultation.

4. Emergency Alert System:

Notifies caregivers or medical contacts during sudden asthma distress.

5. Wearable Health Devices Market:

Can be integrated into smart healthcare and wellness products.

6. Research & Data Analysis:

Helps study respiratory patterns and asthma trigger patterns.

ADVANTAGES:

1. Real-time Monitoring:

Continuously tracks breathing rate, air quality, temperature and humidity.

2. Early Detection of Asthma Triggers:

Identifies harmful environmental conditions before they cause an attack.

3. Instant Alerts:

Sends notifications to the user/caregiver through mobile app or Web-Server.

4. Portable & Wearable:

Lightweight wristband + handheld spirometer makes it easy to use anywhere.

5. Improved Asthma Management:

Users can track trends and maintain better control over their condition.

6. Reduces Emergency Visits:

Helps prevent severe asthma episodes by monitoring continuously

DISADVANTAGES:

1. Requires Smartphone Connectivity:

BLE / WiFi connection or internet is needed for full functionality.

2. Battery Dependence:

Wristband and spirometer need regular charging.

3. Sensor Sensitivity:

Sensors may be affected by moisture, sweat, or improper wearing.

4. User Compliance Required:

The user must wear the device properly and use the spirometer correctly.

5. Environmental Interference:

Gas and temperature readings can be affected by sudden outdoor condition.

CHAPTER 5

Literature Survey

1. Peak Flow Meter with Measurement Analysis (Anisa, Hamzah, Makruf)

Year: 2020

Technology: Custom Peak Flow Meter (PFR) with MPX5100GP sensor, Arduino Nano, SD Card, OLED

Results and Conclusion:

The MPX5100GP-based PFR device provides inexpensive digital measurement but suffers from an inefficient data process; it requires manually moving the data from an SD card to a computer for graphing and proper analysis, preventing immediate clinical assessment.

2. IoT-based Pulmonary Function Monitoring for Children with Bronchial Asthma (Sari Luthfiqah et al.)

Year: 2024

Technology: Spirometer with Flow Turbine Sensor, MAX30102 SpO2 sensor, ESP32, Kodular Android application

Results and Conclusion:

This IoT spirometer achieved good data transmission reliability and reasonable accuracy for the FEV1/FVC ratio. However, the chosen flow turbine sensor yielded unacceptable high error values for absolute FEV1 (+14%) and FVC (+8.7%), demonstrating that the sensing technology used was imprecise for measuring critical volumes and highlighting the need for a more accurate sensor.

3. Smart Monitoring System for Asthma Patients

Year: 2021

Technology: NodeMCU, MQ-2, DHT11 temperature sensor, pulse rate sensor, GPS module, OLED display, Android app, Firebase cloud services

Results and Conclusion:

The reference paper proposed a portable sensor device with app and cloud integration, leaving spirometer functionality as future work. Our project enhances this by integrating a fully functional portable spirometer (MPXV7002DP + ESP32 + battery + charger) with BLE connectivity.

4. Smart Wearable System to Assist Asthma Patients (Niranjana S et al.)

Year: 2021

Technology: Wearable system with gas, temperature, humidity, and pulse rate sensors; NodeMCU (ESP8266), Firebase, Android App

Results and Conclusion:

This wearable gathers environmental trigger data but lacks spirometry capabilities (PEF, FEV1, FVC), meaning it monitors causes but not airway function impairment.

5. IoT Based Smart Inhaler for Context-Aware Service Provisioning (Aninda Dey et al.)

Year: 2020

Technology: Smart inhaler with sensors (SpO2, HR, Gas, Temp/Humidity) integrated with an app

Results and Conclusion:

Provides personalized advice but lacks spirometry measurement, relying only on symptoms/external factors.

6. MAX30102 Photometric Biosensor Coupled to ESP32-Webserver (Uriel Abe Contradi et al.)

Year: 2021

Technology: Wearable finger/wrist device with MAX30102, ESP32 (T-Display), 3D-printed case, webserver

Results and Conclusion:

Provides continuous SpO2 and HR monitoring but lacks spirometry and environmental trigger correlation.

Expert Feedback

- Discussed with two medical professionals (BHMS)
- Both appreciated the idea and confirmed no similar devices currently exist
- They agreed it could significantly benefit asthma patients
- Confirms project usefulness and originality

Consulted Experts:

Dr. Omkar Tarihalkar, BHMS

Dr. Arjun Shirale, BHMS

CHAPTER 6

RESULTS

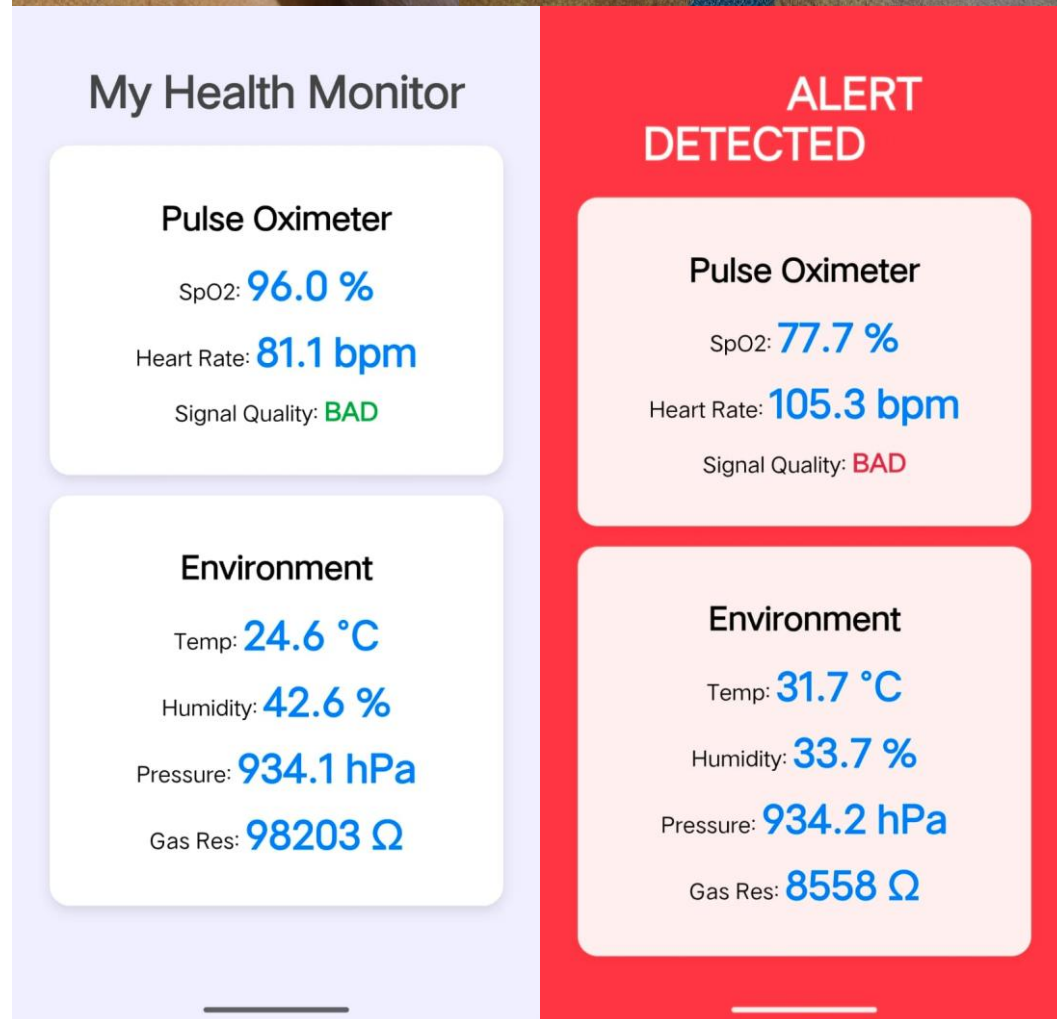
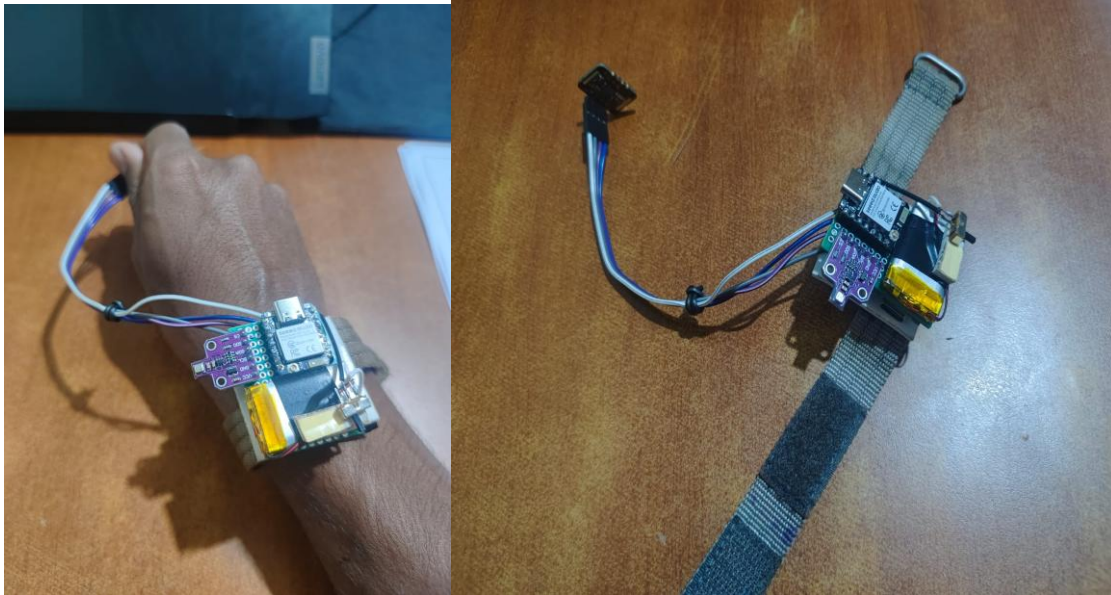
The Smart Asthma Kit was successfully developed and tested using the wristband module, spirometer module, and the mobile application. The system accurately monitored heart rate, SpO₂, temperature, humidity, air quality, and lung airflow in real time. The wristband transmitted physiological and environmental readings to the mobile application through BLE without noticeable delay.

The spirometer module measured airflow values reliably, and the readings responded accurately to variations in exhalation strength. The mobile application displayed all sensor data clearly and generated alert notifications whenever parameters crossed unsafe limits. The BLE connection remained stable during testing, ensuring smooth data transfer between the devices and the app.

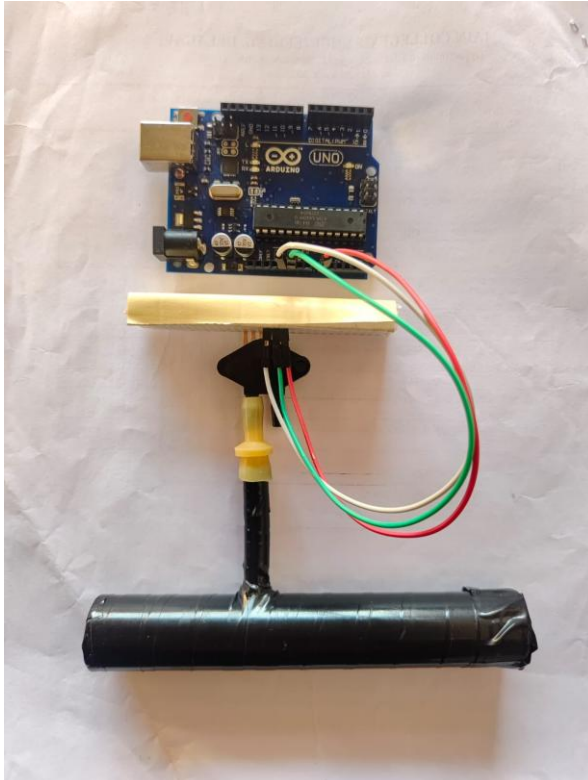
Overall, the results confirmed that the proposed system is capable of continuous monitoring, fast data transmission, and timely alert generation, making it useful for supporting asthma management and early detection of triggering conditions.

Prototype and Output images:

Wristband:



Spirometer:



```
=== Spirometry Test (Demographic grading) ===
You will be prompted for age, sex (M/F), and height (cm).
Commands at any time: k <value> (set K)

Enter age (years) and press ENTER (current 30):
Enter sex (M or F) and press ENTER (current M):
Enter height in cm and press ENTER (current 170):
Using Age=20 Sex=M Height=170 cm

Press ENTER (empty line) to start test.

Get ready:
1) Take a deep breath
2) Seal lips on mouthpiece
3) Blow as hard & fast as you can and keep blowing until told to stop (max 6 s)
3...
2...
1...
Start now!

----- TEST RESULT -----
Duration: 3130 ms
PEF: 431.7 L/min
FEV1: 4.930 L
FVC: 5.339 L
Pred FEV1 (est): 3.800 L
FEV1/Pred: 130 %
Health: Good
K used: 35.000
-----
```

CHAPTER 7

CONCLUSION

The Smart Asthma Kit consisting of a wearable wristband and a portable spirometer was successfully designed and implemented. The system effectively monitored heart rate, SpO₂, air quality, temperature, humidity, and lung airflow in real time. The mobile application provided clear data visualization and timely alerts whenever abnormal conditions were detected. The results show that the system can support asthma patients by helping them track their respiratory health, identify triggers early, and reduce the chances of sudden asthma-related emergencies. Overall, the project demonstrates a reliable and user-friendly solution for continuous asthma management.

FUTURE SCOPE

- The system can be enhanced by integrating cloud storage for long-term data analysis and doctor monitoring.
- GPS-based emergency alerting can be added to notify family members or healthcare providers during severe asthma events.
- AI-based prediction models can be implemented to forecast asthma attacks based on historical data trends.
- The wristband and spirometer can be redesigned into a compact commercial product with improved battery life and durability.

CHAPTER 8

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